SOFTWARE ENGINEERING FUNDAMENTALS • COMPUTER SCIENCE & WEB ENGINEERING

DIGITAL WAITERS: INTRO TO REST APIS

Level: Young Engineers & Tech Enthusiasts **Module:** Foundations of REST APIs

1. UNDERSTANDING REAL-TIME APP DATA

When you open an app like Zomato or Swiggy, it instantly shows live restaurant statuses across the city. Modern applications do not store all this changing data locally on your smartphone; instead, they dynamically request fresh information from remote servers via an **API (Application Programming Interface)**.

2. THE DIGITAL WAITER ANALOGY

How an API Works in Real Life:

- **The Customer (Client / Phone):** You look at options and request specific data/dishes.
- **The Kitchen (Server / Database):** Holds the raw resources, data, and capabilities to fulfill requests.
- **The Waiter (The API):** Takes your order to the kitchen and delivers the response back to your table.

3. WHAT IS AN ENDPOINT?

APIs organize their services using path routes called **endpoints**. Think of entering a department store like Big Bazaar: the Dairy Counter serves milk, while the Electronics Counter serves gadgets. Similarly:

- `/canteen/menu` fetches menu or item data.
- `/canteen/order` handles order submissions.

4. THE 4 BASIC API ACTION VERBS

HTTP VERB	CORE OBJECTIVE	WAITER ANALOGY
GET	Fetch / Read existing data safely	Asking the waiter to show you today's physical menu.
POST	Create / Submit new records	Placing a brand-new food order at the counter register.
PUT	Update / Modify existing data	Changing your order choices before the food is prepared.
DELETE	Erase / Remove records permanently	Canceling your food order completely.

5. JSON & THE REQUEST-RESPONSE CYCLE

Computers exchange structured data using **JSON (JavaScript Object Notation)**, a lightweight text format based on key-value pairs:

```
{
  "restaurantName": "Canteen Express",
  "isOpen": true,
  "availableItems": ["Samosa", "Sandwich", "Juice"]
}
```

The Request-Response Journey: (1) Client builds request → (2) Waiter carries request over network → (3) Server processes & builds response → (4) Waiter brings data back → (5) App displays updated UI!

PRACTICAL LAB & ARCHITECT WORKSHEET

STUDENT WORKSHEET: REST API SYSTEM ARCHITECT

ACTIVITY 1: SCHOOL CANTEEN API BLUEPRINT

Fill out the API Specification table for a School Canteen App system:

TASK / FEATURE	ENDPOINT PATH	HTTP VERB	EXPECTED PAYLOAD / BEHAVIOR
View Daily Menu	`/canteen/menu`	GET	Returns array of available items & prices.
Place Lunch Order	`/canteen/order`	POST	Submits item selection; returns Order ID.
Update Seat Number	`/canteen/order/seat`	PUT	Modifies delivery seat location for active order.
Cancel Active Order	`/canteen/order/cancel`	DELETE	Erases order from kitchen queue.

ACTIVITY 2: CONSTRUCTING JSON DATA MODELS

Write a valid JSON payload snippet representing a student profile requesting lunch. Include key-value pairs for `studentName`, `grade`, `isVegetarian`, and an array `orderedItems`:

```
{
  "studentName": "____________________",
  "grade": ____,
  "isVegetarian": ________,
  "orderedItems": [
    "____________________",
    "____________________"
  ]
}
```

ACTIVITY 3: STEP-BY-STEP CODE WALKTHROUGH (PYTHON REQUESTS)

Complete the code snippets below by filling in the missing HTTP verbs, endpoint URLs, and JSON parameters to simulate real API calls using Python:

Step 3.1: Fetching Menu Data (GET Request)

```python
# Step 1: Import Python library to act as our Digital Waiter
import requests

# Fill in the missing verb and endpoint route to fetch the canteen menu
url = "https://api.schoolcanteen.com/canteen/menu"
response = requests.____________(url) # Hint: Verb for reading data

# Step 2: Convert the JSON response into Python dictionary format
menu_data = response.json()
print("Today's Menu:", menu_data)
```

Step 3.2: Submitting a New Lunch Order (POST Request)

```python
# Step 1: Define payload dictionary for your new order
new_order = {
    "studentName": "Rahul Sharma",
    "item": "Vegetable Sandwich",
    "quantity": 2
}
```

```python
# Step 2: Fill in missing verb and JSON parameter to place the order
order_url = "https://api.schoolcanteen.com/canteen/order"
response = requests.____________(order_url, json=____________)

print("Order Status Code:", response.status_code)
```

Step 3.3: Modifying Seat Location (PUT Request)

```python
# Fill in the HTTP method to update active order details
update_url = "https://api.schoolcanteen.com/canteen/order/104"
updated_info = {"seatNumber": "Table 12B"}

response = requests.____________(update_url, json=updated_info)
```

Step 3.4: Canceling an Active Order (DELETE Request)

```python
# Complete code to send a delete request for order ID 104
cancel_url = "https://api.schoolcanteen.com/canteen/order/104"

response = requests.____________(cancel_url)
print("Cancellation confirmation:", response.status_code)
```

EVALUATION & CONCEPTUAL MASTERY

STUDENT PRACTICE ASSESSMENT

ACTIVITY 4: HTTP VERB MAPPING CHALLENGE

Match each real-world app interaction to its appropriate HTTP verb (GET, POST, PUT, or DELETE):

APP SCENARIO	CORRECT HTTP VERB
1. Reloading your Instagram feed to see new posts	___________
2. Publishing a new photo with a caption	___________
3. Changing your profile bio description	___________
4. Removing an old post from your gallery	___________

SECTION A: MULTIPLE CHOICE QUESTIONS

Q1. In the "Digital Waiter" analogy, what component does the Waiter represent?

[A] The Database Server
[B] The Client / Smartphone User
[C] The Application Programming Interface (API)
[D] The Physical Wi-Fi Router

Q2. Which HTTP verb is used when an application only needs to read or fetch data without making any changes on the server?

[A] POST
[B] GET
[C] PUT
[D] DELETE

Q3. What does JSON stand for in web development?

[A] Java Standard Output Network
[B] JavaScript Object Notation
[C] Joint System Operations Node
[D] Judicial Server Open Syntax

Q4. Which of the following best defines an API Endpoint?

[A] The physical power cable connecting a server to the wall
[B] A structured path route (e.g. /canteen/menu) that targets a specific server resource
[C] The final screen shown when an application crashes
[D] A password used to log into an administrator dashboard

SECTION B: SHORT ANSWER & CONCEPTUAL MASTERY

Q5. Why can't a mobile food delivery app keep all restaurant status data stored directly inside the app download file? Explain what would happen if it did.

Q6. Trace the 5 steps of the Request-Response Cycle when a user taps "Refresh Menu" on their phone screen.

SECTION C: REAL-WORLD SCENARIO PROBLEM

Q7. Scenario: You are building a Ride-Hailing App (like Uber or Ola). A passenger opens the app, views available drivers nearby, requests a ride, updates their drop-off location, and later cancels the ride.

(a) Identify which HTTP verb (GET, POST, PUT, DELETE) is used for each step of the journey:

- View available drivers: ____________________
- Request a ride: ____________________
- Update drop-off location: ____________________
- Cancel the ride: ____________________

(b) Propose a clean endpoint path for requesting a new ride: